

- Q.28 Describe simple gear train with neat diagram. (CO4)
- Q.29 Write any four difference between flywheel and governor. (CO5)
- Q.30 Draw & explain the turning moment diagram for 4-stroke I.C Engine. (CO5)
- Q.31 In a watt governor the length of each arm is 300 mm and they are pivoted on the axis of rotation. Determine the height of governor and radii of rotation of the balls when the speed of governor is 80 rpm. (CO6)
- Q.32 Write various steps to draw displacement diagram when the follower moves with uniform acceleration. (CO7)
- Q.33 Name various types of follower. Explain two types of followers. (CO7)
- Q.34 Discuss the balancing of rotating masses in the same plane. (CO8)
- Q.35 Write any four causes of vibrations in machines. (CO9)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Derive an expression for the ratio of driving tensions for a flat belt drive. (CO3)
- Q.37 The mass of a flywheel is 4000 kg, which has radius of gyration of 2m. Calculate the amount of energy the flywheel will store in changing its speed from 460 rpm to 462 rpm. (CO5)
- Q.38 Explain the construction and working of Porter governor with the help of neat diagram. (CO6)

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5th Sem / Mech, Prod, Mecatronics, (4th sem), CAD/CAM, Mech. Engg. (Fabrication Tech.), Mechanical Engg. (CAD/CAM Design & Robotics)

Subject:- Theory of Machines

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 _____ is that branch of theory of machine which deals with the Forces and their effect while acting upon the machine parts in motion. (CO1)
- a) kinematics b) Dynamics
c) Kinetics d) Statics
- Q.2 The piston and cylinder form a _____. (CO1)
- a) Sliding pair b) Rolling pair
c) Screw pair d) Turning pair
- Q.3 The power transmitted by a belt is maximum when the maximum tension in the belt is equal to (CO2)
- a) T_c b) $2 T_c$
c) $3 T_c$ d) $4 T_c$
- Q.4 When two pulleys of different diameters are connected by means of an open belt drive, then the angle of contact taken into consideration should be of the (CO3)
- a) larger pulley
b) smaller pulley
c) average of two pulleys
d) None of these

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- Q.5 The two parallel and coplanar shafts are connected by gears having teeth parallel to the axis of the shaft. This arrangement is called _____ (CO4)
- a) Spur gearing b) Helical gearing
c) Bevel gearing d) Spiral gearing
- Q.6 Energy is stored in a flywheel by virtue of its _____ (CO5)
- a) Heavy mass
b) Large diameter
c) High speed
d) Heavy mass and slow speed
- Q.7 Which one of the following is Dead weight type governor? (CO6)
- a) Porter Governor
b) Hartnell Governor
c) Wilson-Hartnell Governor
d) Hartung Governor
- Q.8 The cam follower generally used in automobile industry is _____ (CO7)
- a) knife edge follower
b) Flat faced follower
c) Spherical face follower
d) Roller follower
- Q.9 The balancing of rotating and reciprocating parts of an engine is necessary when it runs at _____ (CO8)
- a) Slow speed b) medium speed
c) high speed d) constant speed
- Q.10 When the body vibrates under the influence of external force, then the body is said to be under _____ (CO9)
- a) free vibration b) natural vibrations
c) damped vibrations d) forced vibrations

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Define the following:

- Q.11 Theory of Machine. (CO1)
Q.12 Kinematic pair. (CO1)
Q.13 Structure (CO2)
Q.14 Gear train. (CO3)
Q.15 Turning moment diagram. (CO5)
Q.16 Fluctuation of speed (CO5)
Q.17 Stability of governor. (CO6)
Q.18 Constrained Motion. (CO7)
Q.19 Dynamic balancing. (CO8)
Q.20 Transverse vibration (CO9)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain various types of kinematic chain. (CO1)
Q.22 Explain Scotch Yoke Mechanism with suitable diagram. (CO1)
Q.23 Write any four difference between Machine and structure. (CO1)
Q.24 Enlist various modes of power transmission. Explain anyone with suitable diagram. (CO2)
Q.25 Find the power transmitted by a belt moving with a velocity of 6.28 m/s. the coefficient of friction between the belt and the pulley is 0.25, angle of lap 160 degree and maximum tension in the belt is 2500 N. (CO2)
Q.26 Explain the concept of crowning of pulleys. (Co3)
Q.27 Explain crank slotted lever quick return motion mechanism.